Customer Behavior Analysis

681M	50K	250K	2.73K	0.20	751K
Total Revinue	Total Customers	Total Orders	Average Purchase	Churn Rate (%)	Total Quantity Sold

gender

Female | Male

payment_method

All

purchase_month

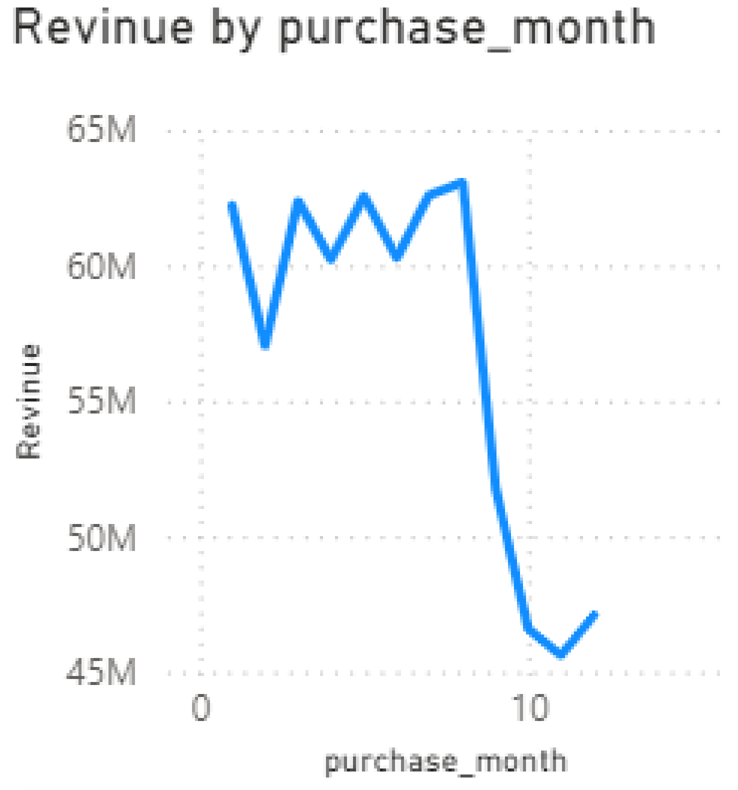
1 | 12

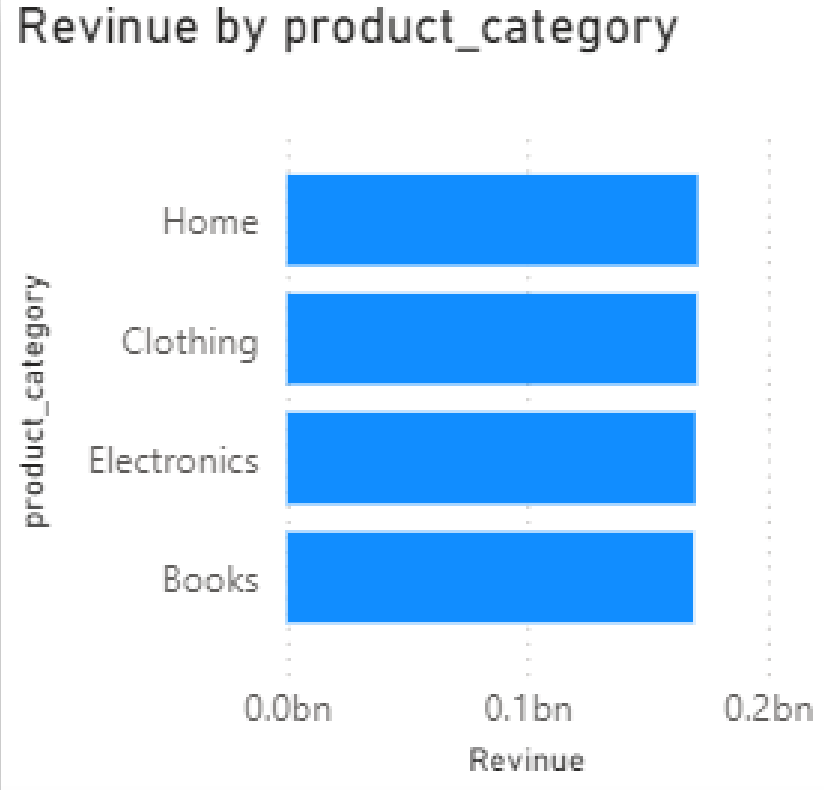
product_category

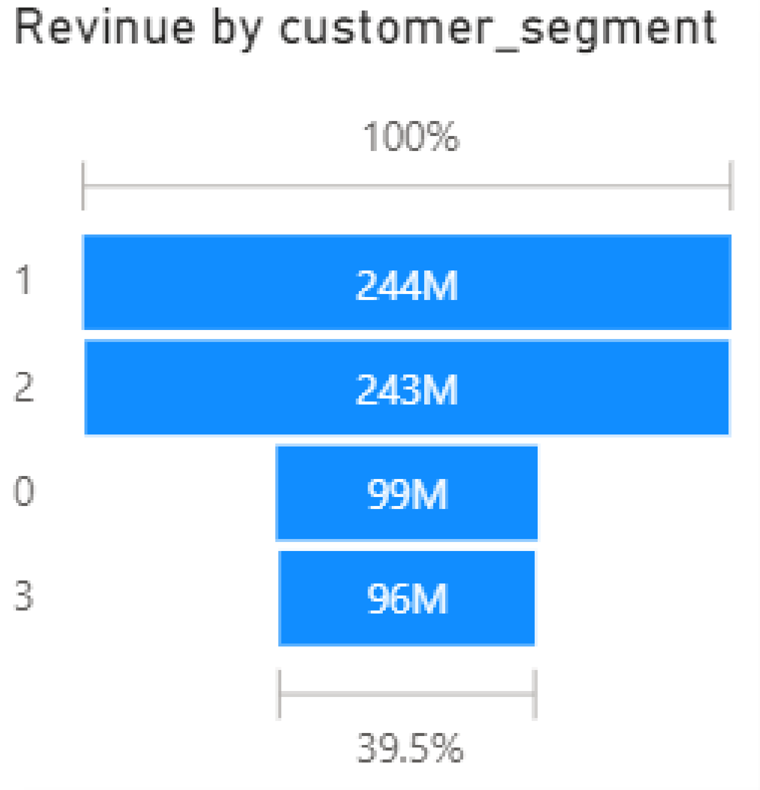
- [ ] Books
- [ ] Clothing

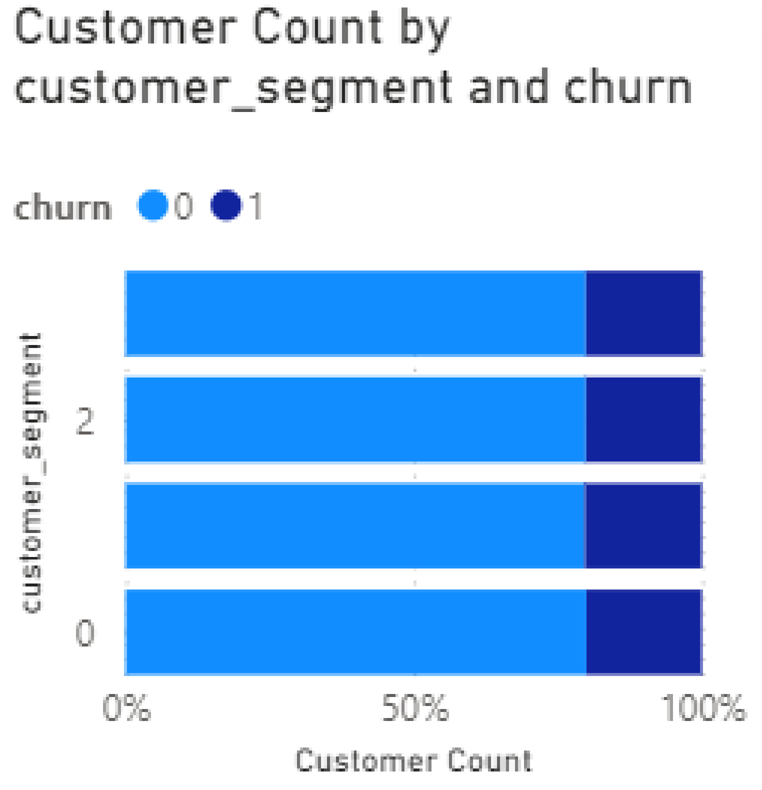
customer_segment

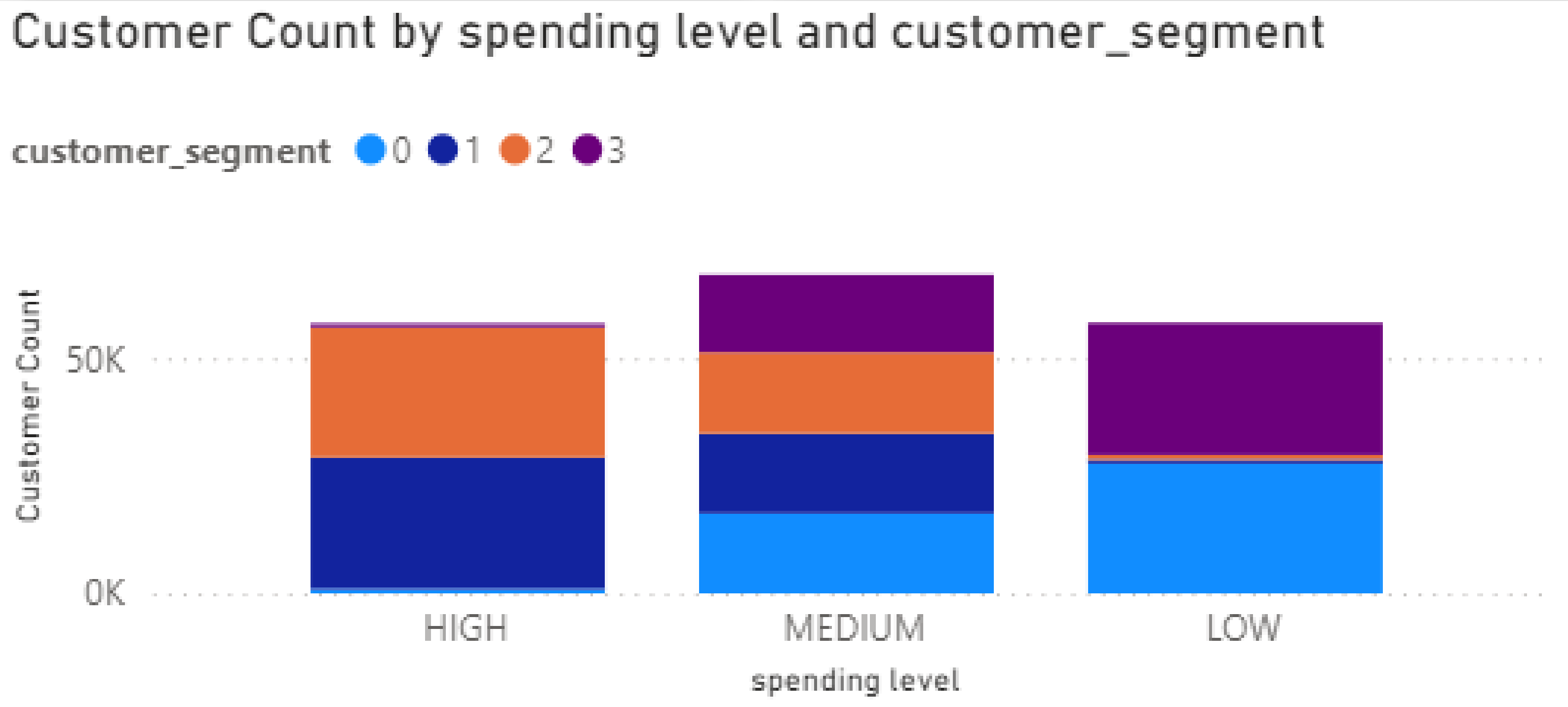
0 | 3

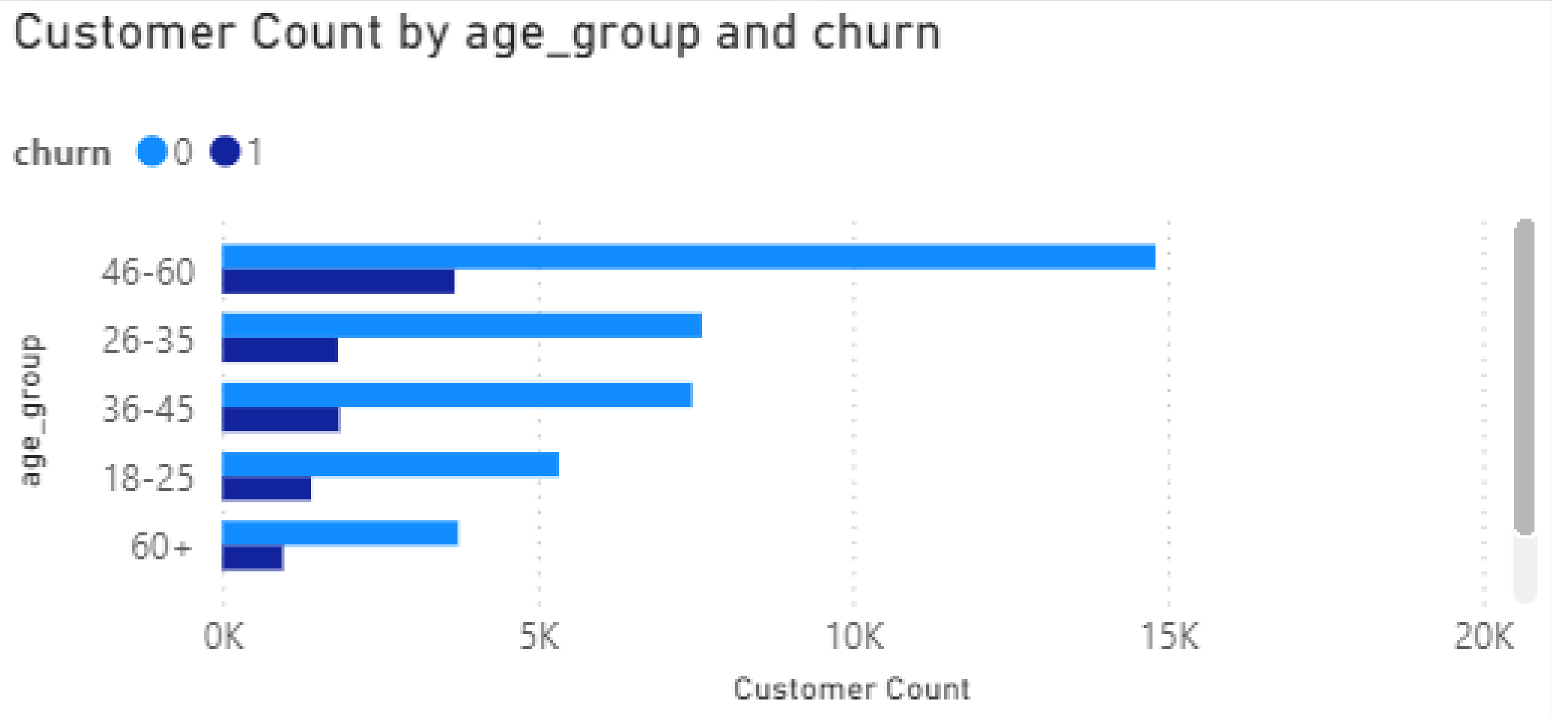


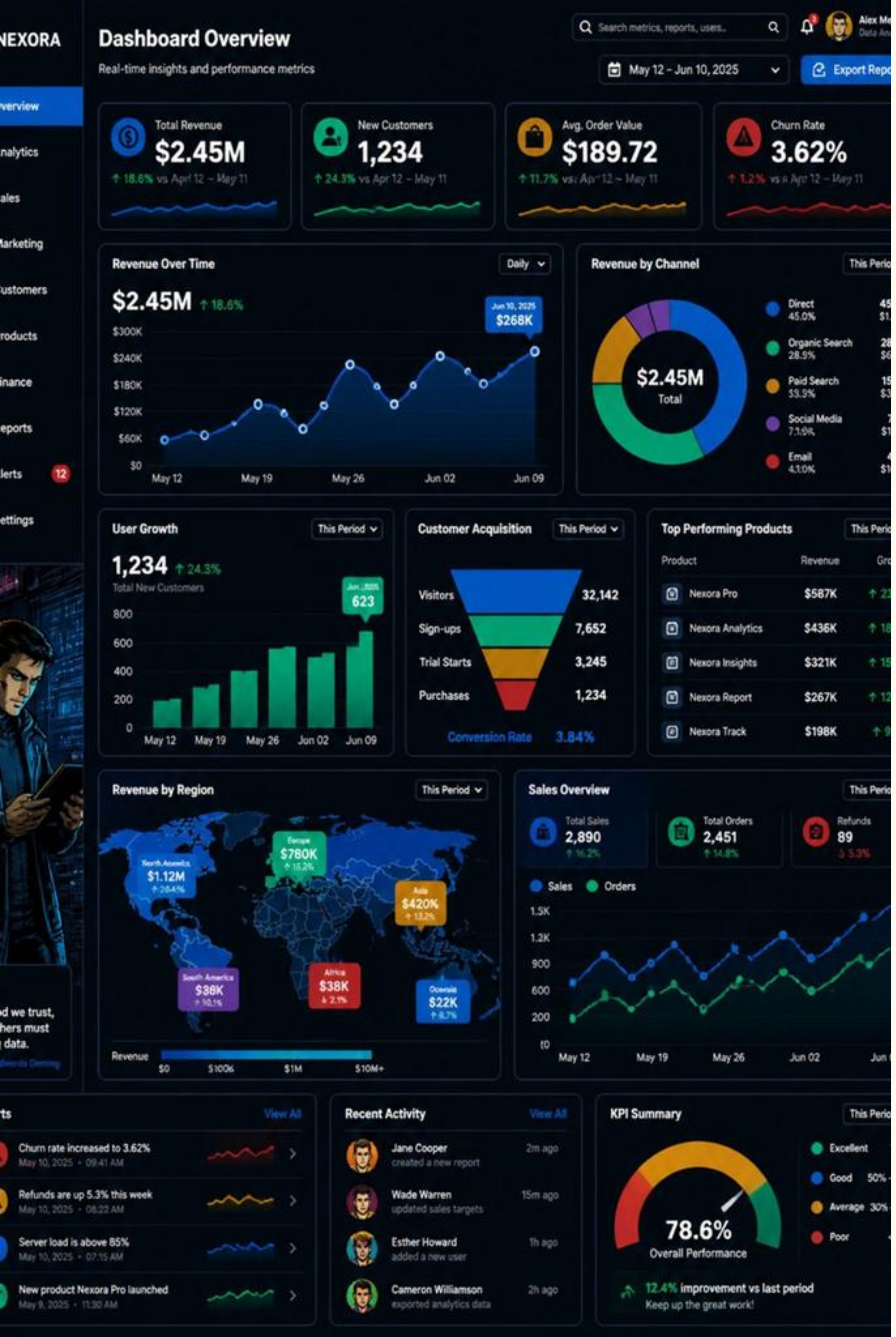












Customer Behavior Analysis

Analyzing Customer Patterns, Purchasing Behavior & Business Insights

Presented By: Chintu Kumar · IIMT College of Engineering, Greater Noida

Internship: Alfido Tech · **Mentor:** Akash Dubey · **Domain:** Data Analytics

PROJECT OVERVIEW

Project Overview & Objectives

This project analyzes customer transaction data to uncover purchasing behavior, revenue drivers, segmentation patterns, churn trends, and actionable business insights.

- Customer purchasing behavior & spending patterns
- Revenue and product category performance
- Customer segmentation & churn identification
- Interactive Power BI dashboard for stakeholders

Exploratory Analysis

Run SQL queries and visualize patterns.

Segmentation & Modeling

Build customer segments and churn models.

DATASET & TOOLS

Dataset & Technologies

250K
Transactions
Total orders in dataset

49.6K
Customers
Unique customer profiles

₹681M
Total Revenue
Aggregate sales value

₹2,725
Avg Purchase
Per transaction value

751K
Units Sold
Total quantity sold

Key features: Customer ID, Purchase Date, Product Category, Price, Quantity, Payment Method, Age, Gender, Returns, and Churn.

Python Stack
Pandas, NumPy, Matplotlib, Seaborn

SQL / MySQL
Business queries & aggregations

Power BI
Interactive dashboards & visuals

Scikit-learn
Segmentation & churn modeling

DATA PREPARATION

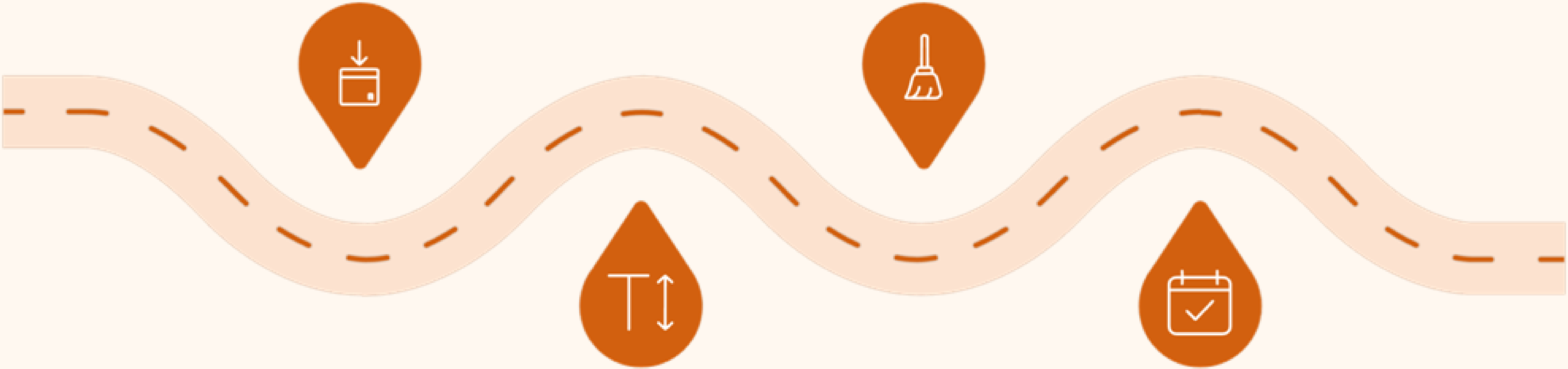
Data Cleaning & Feature Engineering

Import

Load raw dataset for processing

Clean

Check missing values and duplicates



Standardize

Normalize column names to lowercase

Verify

Confirm types and convert dates

A systematic six-step pipeline ensures raw transaction data is clean, consistent, and enriched for downstream analysis.

Data Cleaning

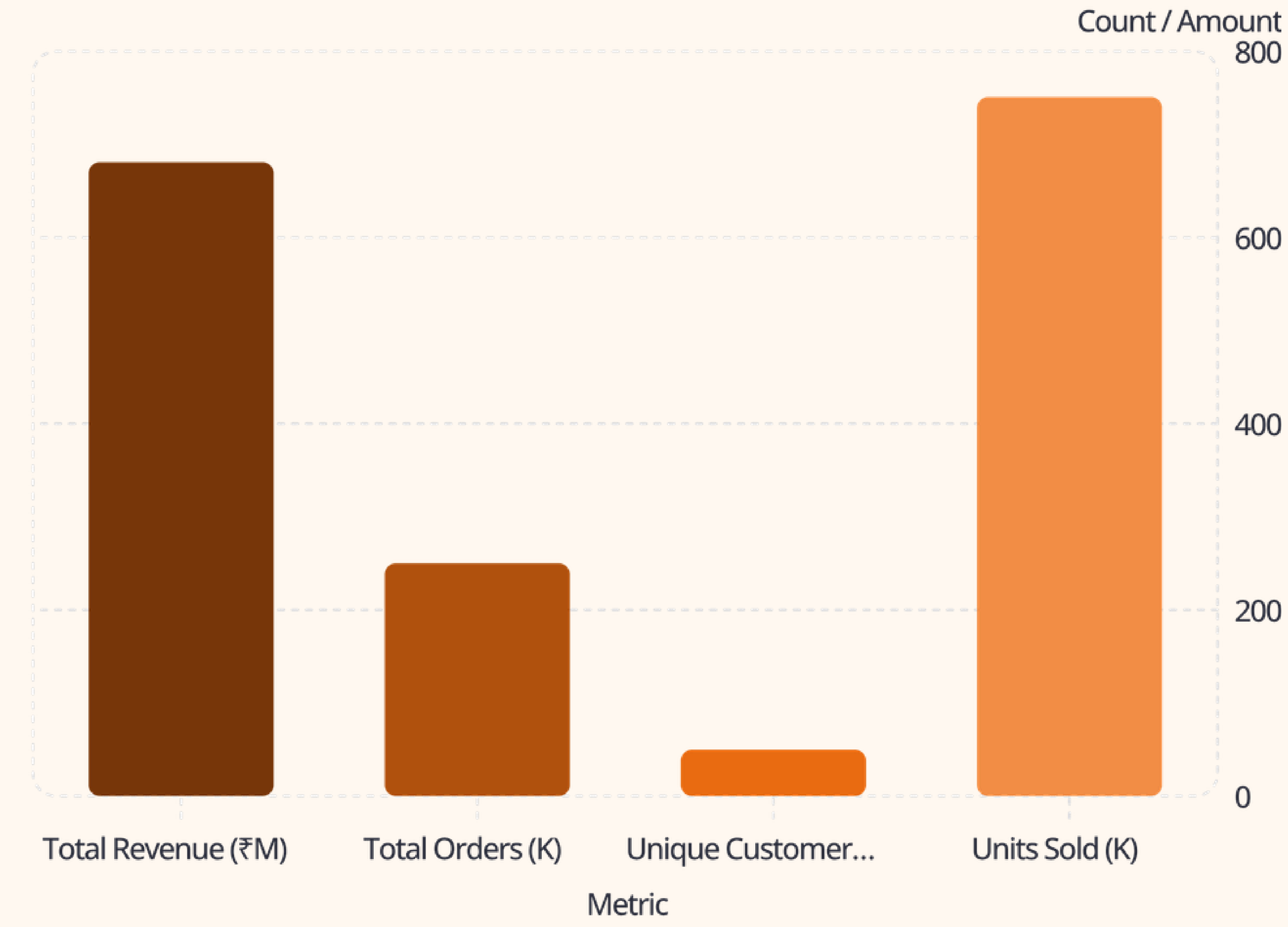
- Standardized column names (lowercase, underscores)
- Checked and handled missing values
- Removed duplicate records
- Verified data types; converted purchase_date to datetime

Feature Engineering

- Derived **Purchase Year, Month, Day** for temporal analysis
- Created **Purchase Quarter** for seasonal trend analysis
- Enabled time-based filtering and period comparisons

ANALYSIS

EDA & SQL Business Analysis



Exploratory Data Analysis

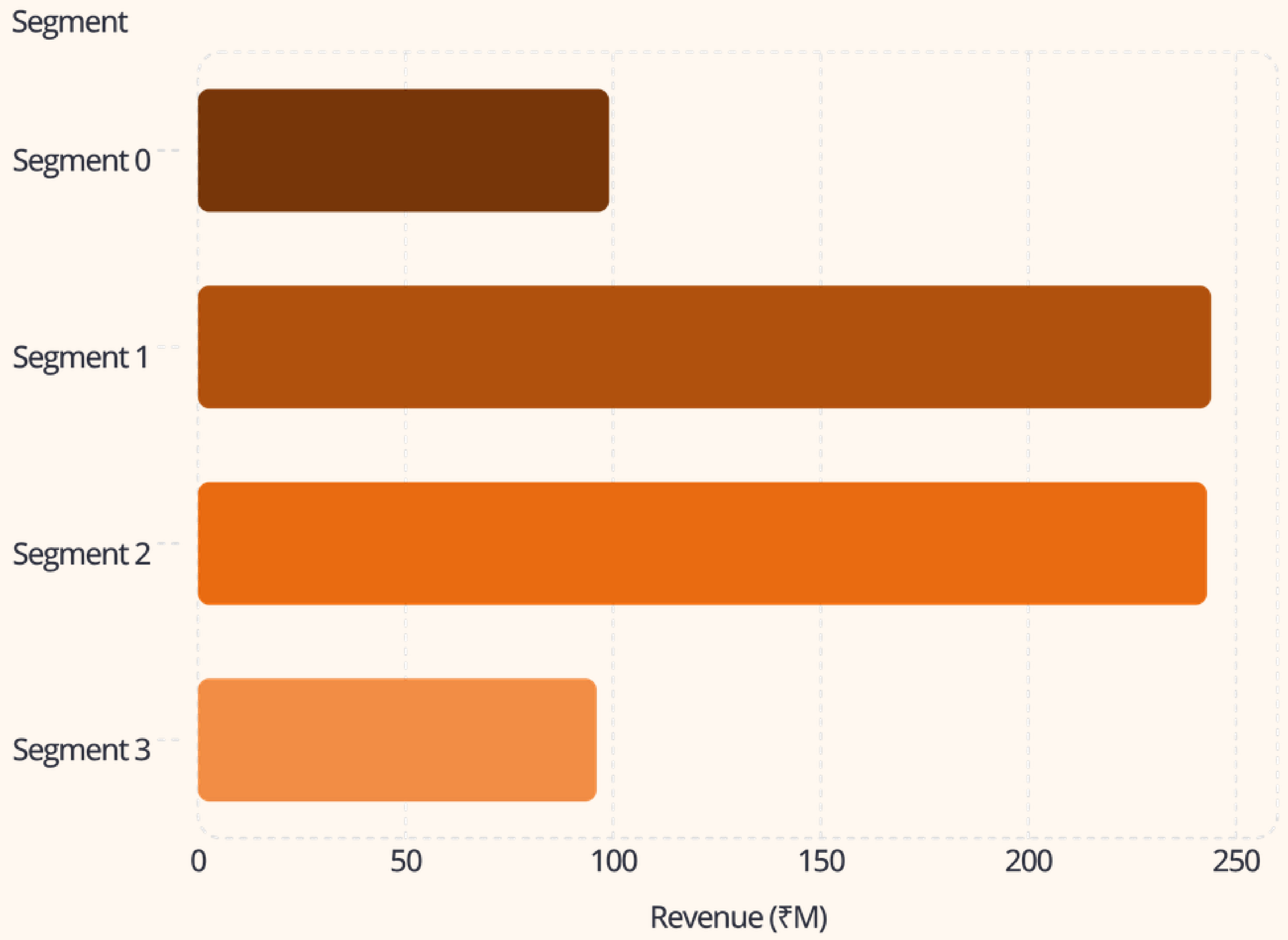
EDA examined revenue trends, customer behavior, product categories, monthly patterns, payment methods, and spending levels across the dataset.

SQL Business Queries

- Customer & revenue analysis
- Product performance by category
- Spending level classification
- Segment-wise aggregations
- Churn rate computation

SEGMENTATION & CHURN

Customer Segmentation & Churn Analysis



Segment Insights

Segments 1 and 2 are the highest revenue contributors, together accounting for nearly **72% of total revenue**. Customers were also classified as Low, Medium, or High spenders.

Churn Analysis

- Overall churn rate: **20%**
- Analyzed by segment and age group
- Churn patterns inform retention strategy

DASHBOARD

Interactive Power BI Dashboard

₹681M Revenue

Total sales value

50K Customers

Unique profiles

250K Orders

Transactions recorded

₹2.73K Avg

Per purchase value

20% Churn

Overall churn rate

751K Units

Total quantity sold

Visualizations: Revenue by Month · Revenue by Category · Revenue by Segment · Spending Levels · Churn by Segment · Churn by Age Group

Filters: Gender · Payment Method · Purchase Month · Product Category · Customer Segment

FINDINGS

Key Business Findings

1

₹681M Revenue

From 250K transactions across 49.6K customers

2

Segments 1 & 2 Lead

Contribute ~72% of total revenue

3

Home & Clothing

Among the strongest-performing categories

4

46–60 Age Group

Highest customer count; key demographic

5

20% Churn Rate

Signals a significant retention opportunity

6

₹2,725 Avg Purchase

751K units sold; spending varies by segment

RECOMMENDATIONS

Recommendations & Future Scope

Actionable Recommendations

- Loyalty rewards for high-value customers
- Personalized campaigns to reduce 20% churn
- Cross-selling to Low & Medium spenders
- Promote Home & Clothing categories
- Seasonal campaigns during weak revenue months
- Segment-based personalized marketing

Future Scope

- ML-based churn prediction models
- K-Means clustering for refined segmentation
- Customer Lifetime Value (CLV) modeling
- Product recommendation engine
- Sales forecasting with time-series models
- Real-time dashboard with automated updates

Targeted Recommendations

Better Retention

Higher Revenue

Implementing these recommendations creates a virtuous cycle of retention and revenue growth.

THANK YOU

Thank You

This project demonstrates the end-to-end application of data analytics — from raw transaction data to strategic business insights — using Python, SQL, and Power BI.

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